SAGI Summer School 2025: Introduction to Fourier Analysis

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Fourier Transform

1. We have now seen that we can represent a periodic function $x(t)$ as a summation of harmonically related complex exponentials:

$$x(t) = \sum_{n=-\infty}^{\infty} a_n e^{in\omega_0 t} \tag{1}$$

where $\omega_0 = \frac{2\pi}{T}$ and T is the fundamental period of the function $x(t)$.

2. Such a representation is useful because for any **Linear Time-Invariant** (LTI) system we have the following output corresponding to this input:

$$y(t) = \sum_{n=-\infty}^{\infty} H(n\omega_0) e^{in\omega_0 t} \tag{2}$$

where $H(\omega)$ is the complex valued **transfer function** of the LTI system.

Fourier Transform

1. After some work, we showed that the Fourier series coefficients can be found from the input signal $x(t)$ with:

$$a_k = \frac{1}{T}\int_{t}^{t+T} x(t)e^{-ik\omega_0 t} \tag{3}$$

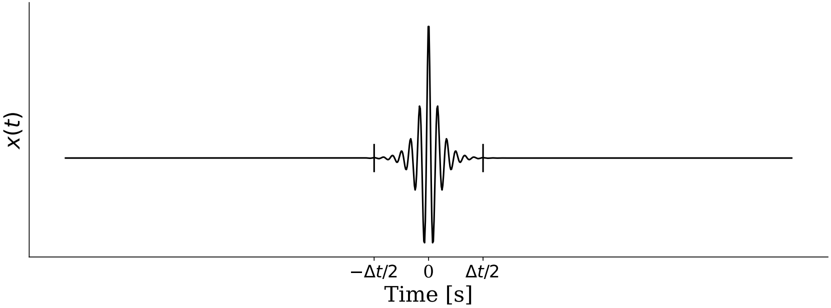
2. We used this expression to show for the square wave:

$$a_0 = \frac{1}{2} \tag{4}$$

$$a_k = \frac{\sin(k\pi/2)}{k\pi},\ k \neq 0 \tag{5}$$

Fourier Transform

1. To come up with equation 3 for the Fourier Series coefficients, we had to assume that our input signal $x(t)$ was periodic.
2. Can we write a representation similar to equation 3 for signals which are not periodic? Yes, and the result is known as the **Fourier Transform**.
3. Suppose we have a signal $x(t)$ which is non-zero for only a finite time window. This signal might correspond to say, a fast-radio burst, and could look something like:



Fourier Transform

❶ Let's construct a new signal $\tilde{x}(t)$ that is a periodic version of the original signal $x(t)$. This is achieved by copying the original signal at regular time intervals T:

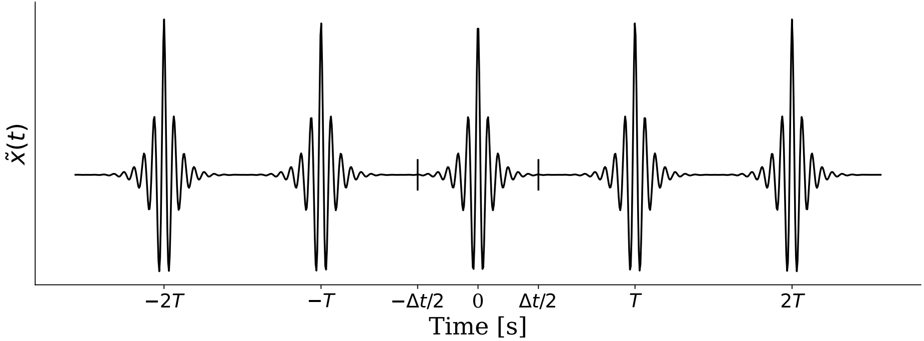


Figure: $\tilde{x}(t)$, constructed by copying $x(t)$ at regular time intervals. The copies extend out to $\pm\infty$

Fourier Transform

1. Because $\tilde{x}(t)$ is periodic with fundamental period T, we can use equation 3 to determine it's Fourier Series coefficients:

$$a_k = \frac{1}{T}\int_{-\Delta t/2}^{-\Delta t/2+T} \tilde{x}(t)e^{-ik\omega_0 t}dt \tag{6}$$

where $\omega_0 = 2\pi/T$

2. Because $\tilde{x}(t) = 0$ for $\Delta t/2 < t < T - \Delta t/2$, we can rewrite the bounds of integration:

$$a_k = \frac{1}{T}\int_{-\Delta t/2}^{\Delta t/2} \tilde{x}(t)e^{-ik\omega_0 t}dt \tag{7}$$

Fourier Transform

1. Now, suppose we take the period $T \to \infty$. In this limit, we have $\tilde{x}(t) = 0$ for all $|t| > \Delta t/2$ so that:

$$a_k = \frac{1}{T} \int_{-\infty}^{\infty} \tilde{x}(t) e^{-ik\omega_0 t} dt \tag{8}$$

2. Now define a function continuous in the frequency variable ω:

$$X(\omega) = \int_{-\infty}^{\infty} \tilde{x}(t) e^{-i\omega t} dt \tag{9}$$

3. Using this definition in equation 8:

$$a_k = \frac{1}{T} X(k\omega_0) \tag{10}$$

Fourier Transform

1. Now, substituting these coefficients into equation 1 for $\tilde{x}(t)$:

$$\tilde{x}(t) = \frac{1}{T} \sum_{n--\infty}^{\infty} X(k\omega_0) e^{ik\omega_0 t} \tag{11}$$

$$= \frac{1}{2\pi} \sum_{n=-\infty}^{\infty} X(k\omega_0) e^{ik\omega_0 t} \omega_0 \tag{12}$$

where the second equality uses $\omega_0 = \frac{2\pi}{T}$.

2. As we take $T \to \infty$, we have: $\tilde{x}(t) \to x(t)$, $k\omega_0 \to \omega$, $\omega_0 \to d\omega$, $\sum \to \int$ and equation 12 becomes:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{i\omega t} d\omega \tag{13}$$

Fourier Transform

Result: Fourier Transform Equations

Equations 9 and 13 are the Fourier Transform equations, which relate a general aperiodic signal $x(t)$ with it's frequency components $X(\omega)$. The importance of these equations in nearly every field of experimental science cannot be overstated. I repeat them here for emphasis:

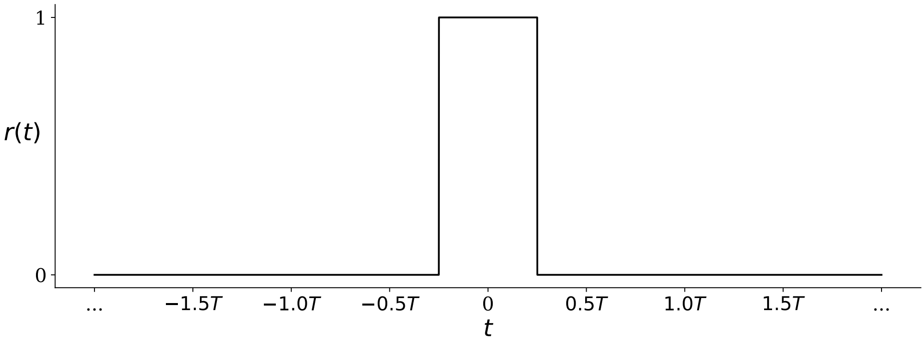
$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{i\omega t}d\omega \tag{14}$$

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-i\omega t}dt \tag{15}$$

Rectangle Signal

❶ Let's apply these equations to an example function, the rectangle signal:

$$r(t) = \begin{cases} 1, & |t| \leq \frac{T}{4} \\ 0, & \frac{T}{4} < |t| \end{cases} \tag{16}$$



Rectangle Signal

1. Notice that the square wave can serve as the periodic version, $\tilde{x}(t)$, of the rectangle signal. Perhaps we can expect the Fourier Series coefficients for the square to be related to the Fourier Transform of the rectangle.
2. The Fourier Transform of the rectangle is fairly straightforward to find:

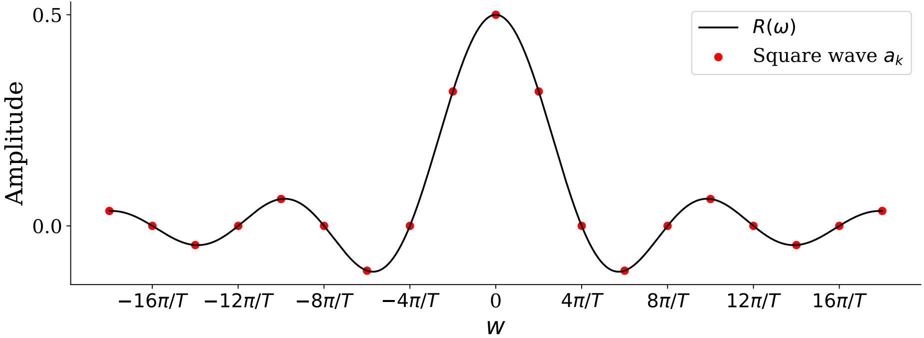
$$R(\omega) = \int_{-T/4}^{T/4} e^{-i\omega t} dt = \left. \frac{e^{-i\omega t}}{-i\omega} \right|_{-T/4}^{T/4} = \frac{e^{i\omega T/4} - e^{-i\omega T/4}}{i\omega} \tag{17}$$

3. Using $(e^{ix} - e^{-ix})/2i = \sin(x)$:

$$R(\omega) = \frac{2\sin(\frac{\omega T}{4})}{\omega} \tag{18}$$

Rectangle Signal

1. Plotting $R(\omega)$ (black line) along with the Fourier Series coefficients for the square wave (red points) we see that the square wave Fourier Series exactly samples the Fourier Transform of the rectangle signal!



Rectangle Signal

1. There are a few other interesting things to note about $R(\omega)$. First, we see that it is completely real-valued. This means that the phase is 0 for all values of ω, i.e. $\angle R(\omega) = 0$.
2. If we were to displace the original rectangle $r(t)$ so that it is not completely symmetric about the origin, the Fourier Transform $R(\omega)$ will pick up a uniform phase. That is, $R(\omega)$ will have an imaginary component such that $\angle R(\omega) =$ non-zero constant. This constant will be the same for all values of ω.
3. The same is true for the square wave. In our example, we defined a square wave that is completely symmetric about the origin. If we displace the square so that it is not symmetric, each a_k picks up an imaginary component such that $\angle a_k =$ non-zero constant. This constant will be the same for all values of k.

Rectangle Signal

1. The other interesting thing to note is that the form $R(\omega)$ hints at diffraction patterns in optical systems.
2. An aperture on a telescope serves as a type of rectangular function. Electromagnetic radiation within the aperture is allowed to pass into the optics, while radiation outside of the aperture is completely blocked.
3. The distribution of electromagnetic fields far away from a slit (aperture) is related to the field distribution at the slit through the Fourier transform. Because the aperture is a type of rectangular function, when we Fourier transform the fields at the aperture, we get a $\sin(\omega)/\omega$ form of the fields far from the slit which corresponds to the diffraction pattern!